MAL Omics — Proteomics Analysis Report

COPDGene — SomaScan (∼4,979 proteins) — $n = 1{,}306$

Goal. (1) Understand biological correlates of MAL detectable in peripheral plasma. (2) Develop a blood-based proteomic risk score to enrich studies for MAL and emphysema progressors.

Outcome. Emphysema progression = change in CT lung density Phase 3 − Phase 2 (`Change_lung_density_vnb_P2` HU). Negative = worsened.

Table 1 — Demographics

Study population after 3-way merge (proteomics × phenotype × MAL) and complete-case filtering.

Variables not found in the phenotype file:

- *Not found: FEV1pp_P2 (FEV_1% predicted)* — column not present under any searched pattern (`FEV1pp`, `FEV1pct`, `FEV1_percent_pred`). Row left blank.
- *Not found: BMI* — no column matching `^[Bb][Mm][Ii]` found in the phenotype file. Row left blank.

Table 1. Study population (n = 1,306). Mean (SD) for continuous; n (%) for categorical. Emphysema progression = $\rho_{P3} - \rho_{P2}$ (HU).

Characteristic	Overall (n = 1,306)
Demographics	
Age (years)	[from notebook output]
Sex, female	[from notebook output]
Race, White	[from notebook output]
Smoking	
Current smoker (`SmokCigNow_P2`)	[from notebook output]
Pack-years (`ATS_PackYears_P2`)	[from notebook output]
Pulmonary function	
FEV_1% predicted	*Not found: column not found*
FEV_1/FVC (`FEV1_FVC_post_P2`)	[from notebook output]
CT imaging	
Lung density Phase 2 (`lung_density_vnb_P2`, HU)	[from notebook output]
MAL score (`meanmal`)	[from notebook output]
Emphysema progression (HU)	[from notebook output]
Anthropometrics	
BMI (kg/m^2)	*Not found: column not found*

Note. The exact values for rows marked "[from notebook output]" are rendered as an HTML table in cell 9 of `MAL_omics_proteomics.ipynb`. They could not be automatically extracted here because the data are PHI-protected and not committed to the repository.

Figure 1 — Study Design

Not automatically generated. Requires a manual schematic (e.g., BioRender) illustrating the analysis flow: SomaScan proteomics → differential expression → GSEA / STRING → adaptive LASSO PRS → emphysema progression association → causal mediation.

Figure 2 — Protein Expression Landscape of MAL

A. Volcano Plot (protein ∼ MAL)

For each of 4,979 proteins, the model is:

$$\log_2(\text{protein}) \sim \beta_{\text{MAL}} \cdot \texttt{meanmal} + \text{age} + \text{sex} + \text{race} + \text{pack-years} + \text{smoking} + \text{scanner} + \text{PC1–PC5}$$

$\hat{\beta}_{\text{MAL}}$ tells us: for a one-unit increase in MAL, how much does the $\log_2$ protein level change, after adjusting for the listed covariates? FDR correction (Benjamini–Hochberg) across all 4,979 tests.

Each point is one protein. x-axis: MAL $\hat{\beta}$. y-axis: $-\log_{10}(\text{FDR})$. Red = FDR < 0.05 and enriched; blue = FDR < 0.05 and depleted; grey = not significant. Top 20 by FDR are labelled.

Key result. 45 of 4,979 proteins were significant at FDR < 0.05.

Top enriched (higher with more MAL): *ADIPOQ* (Adiponectin, $\beta = 0.27$), *TNNT2* (Troponin T, $\beta = 0.19$), *ADAMTS5*, *NCAM1*, *PAK7*.

Top depleted (lower with more MAL): *ACY1* ($\beta = -0.26$), *ADH4* ($\beta = -0.25$), *RTN4R* (Nogo Receptor), *RET*, *PRSS8*, *SFRP4*.

B. GSEA Barplot

Proteins ranked by MAL t-statistic; gene sets from Hallmark, KEGG, and Reactome. 9 pathways reached adjusted $p < 0.05$.

GSEA significant pathways. NES < 0 = depleted in high-MAL subjects.

Collection	Pathway	NES	padj
Hallmark	Fatty Acid Metabolism	−1.76	0.021
Hallmark	Oxidative Phosphorylation	−1.78	0.021
KEGG	Glycolysis / Gluconeogenesis	−2.02	0.018
Reactome	Biological Oxidations	−2.05	0.007
Reactome	Metabolism of Amino Acids & Derivatives	−1.84	0.013
Reactome	Diseases of Metabolism	−1.76	0.032
Reactome	Phase I Functionalization of Compounds	−1.97	0.037
Reactome	Chemokine Receptors Bind Chemokines	+2.02	0.024
Reactome	Antimicrobial Peptides	+1.94	0.032

Key result. High MAL is associated with broad suppression of energy metabolism and upregulation of chemokine and innate immune signalling.

C. STRING Network

45 FDR-significant proteins mapped to STRING v11.5 (human, interaction score ≥ 400). Network image saved as `Figure2C_STRING_network.png` and embedded in notebook cell 15. No PDF version was generated (only PNG).

Figure 3 — Quartile Plot of Proteomic Risk Score

x-axis: PRS quartile (1 = lowest, 4 = highest). y-axis: $\hat{\beta}$ coefficient (HU/SD) from OLS of emphysema progression on PRS within each quartile, with 95% CI. Dashed line at zero.

Key result. Effect sizes across quartiles are small and confidence intervals cross zero in all quartiles, consistent with the non-significant overall PRS association ($p = 0.079$).

Proteomic Risk Score — Methods and Results

Data split. 70/30 train/test (`set.seed(2024)`), $n_{\text{train}} = 914$, $n_{\text{test}} = 392$.

Step 1 — Log2 transform all protein RFU values:

$$\tilde{x} = \log_2\!\big(\max(x,\ 10^{-6})\big)$$

Transforms skewed RFU values to an approximately normal scale. The 10^{-6} floor prevents $\log(0)$ errors.

Step 2 — Adaptive weights from ridge regression:

$$w_j = \frac{1}{|\hat{\beta}_j^{\text{ridge}}|}$$

A ridge regression is first fit on the residualised MAL outcome (after projecting out the unpenalised covariates). Each protein j gets a weight inversely proportional to its ridge coefficient. Proteins that ridge already identified as informative get *lower* penalty in the LASSO step; irrelevant proteins get *higher* penalty. This gives the adaptive LASSO its oracle property — it tends to select the true model.

Step 3 — Adaptive LASSO selection:

$$\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \sum_j w_j |\beta_j|$$

λ is chosen by 10-fold cross-validation (λ_{min}). Covariates (age, sex, race, smoking, pack-years, scanner, PC1–PC5) are assigned $w_j = 0$ so they are never penalised.

Step 4 — Post-selection OLS refit. The selected proteins are refit by ordinary least squares. LASSO shrinks coefficients toward zero, so the LASSO $\hat{\beta}$ values are biased. OLS on the selected set gives unbiased estimates, proper standard errors, confidence intervals, and p-values.

Step 5 — Compute the risk score for each subject:

$$\text{PRS}_i = \sum_{j \in \mathcal{S}} \hat{\beta}_j^{\text{OLS}} \cdot \tilde{x}_{ij}$$

where $\mathcal{S}$ is the selected protein set. The PRS is then standardised to mean 0, SD 1 across the full cohort so that the regression coefficient has units of HU per SD of the score.

Result. 378 proteins selected. Test $R^2 = 0.157$.

Top 20 PRS proteins by $|\hat{\beta}|$ (OLS).

Gene	Target	$\hat{\beta}$	95% CI (low)	95% CI (high)	p
ZPBP	ZPBP1	+0.286	0.160	0.412	< 0.001
CFI	Factor I	−0.226	−0.398	−0.054	0.010
GREM2	GREM2	+0.193	0.113	0.273	< 0.001
C7	Complement C7	+0.169	0.030	0.309	0.017
SEC13	SEC13	−0.153	−0.254	−0.053	0.003
SERPINF2	α_2-Antiplasmin	+0.149	0.011	0.287	0.035
FSTL1	FSTL1	−0.147	−0.309	0.015	0.075
APOH	β_2-Glycoprotein I	+0.143	−0.049	0.335	0.145
COL3A1	Collagen Type III	+0.132	0.061	0.204	< 0.001
C3	C3a	+0.131	0.021	0.242	0.020
SMS	Spermine Synthase	−0.125	−0.179	−0.072	< 0.001
MATN2	MATN2	−0.125	−0.242	−0.008	0.036
GALNT11	GLT11	−0.122	−0.225	−0.019	0.021
ROBO2	ROBO2	−0.119	−0.220	−0.017	0.022
SEZ6L	SEZ6L	+0.117	0.020	0.214	0.019
FBLN1	Fibulin-1	−0.113	−0.278	0.051	0.175
ADAMTS5	ADAMTS-5	+0.113	0.062	0.163	< 0.001
CCL23	MPIF-1	−0.112	−0.160	−0.063	< 0.001
CREB3L4	CR3L4	+0.108	0.047	0.169	< 0.001
CNTN1	Contactin-1	−0.107	−0.229	0.015	0.086

AIC Comparison — PRS vs. Clinical Model

Three models fitted by OLS for emphysema progression. AIC improvement of ≤ -2 considered meaningful.

Note: BMI was not available in the phenotype file and is absent from the clinical model. This should be revisited when BMI data are available.

AIC comparison.

Model	AIC	ΔAIC vs. Clinical
Clinical (age, sex, race, smoking, pack-years, scanner)	8,815	0
PRS only (PRS + baseline density + scanner)	8,848	+33.1
Clinical + PRS	8,816	+0.8

PRS $\hat{\beta} = 0.42$ HU/SD, 95% CI $[-0.05,\ 0.88]$, $p = 0.079$ in the combined model.

Key result. The PRS does not improve model fit over clinical variables alone (ΔAIC = +0.8, far from the −2 threshold). The PRS-only model is substantially worse than clinical alone (ΔAIC = +33).

Table 2 — Causal Mediation

Pathway tested: MAL → Proteomic Risk Score → Emphysema Progression. Fitted using the `medflex` R package (natural effects framework) with robust sandwich standard errors.

The idea is to decompose the total effect of MAL on progression into:

- **NDE** — how much of the MAL effect operates through pathways *other than* the protein score (the direct path)
- **NIE** — how much operates *through* the protein score (the mediated path)

Two models are fitted internally by `medflex`:

$$\text{Imputation model: } \Delta\rho \sim \texttt{meanmal} \times \text{PRS} + \text{covariates}$$

$$\text{Natural effects model: } \Delta\rho \sim \texttt{meanmal}_0 + \texttt{meanmal}_1 + \text{covariates}$$

$\texttt{meanmal}_0$ is the MAL value at which the mediator (PRS) is set (direct path); $\texttt{meanmal}_1$ is the actual MAL value driving the outcome (indirect path). The difference between them isolates the mediated portion.

Table 2. Natural effects model results.

Effect	Estimate (HU)	z	p
Natural Direct Effect (NDE)	+19.72	2.92	0.004
Natural Indirect Effect (NIE)	−0.12	−0.04	0.971
Total effect	+19.60	—	—
Proportion mediated	−0.6%		

Key result. MAL has a large, significant direct effect on emphysema progression (NDE $p = 0.004$). The proteomic risk score mediates essentially none of this effect (NIE $p = 0.97$, proportion $= -0.6\%$).

Supplement

Supplementary Tables

- **S1 — All differential expression results** (4,979 proteins): gene, target, $\hat{\beta}$, SE, t, p, FDR.
  File: `MAL_omics/mal_de_results.csv`
- **S2 — PRS weights** (378 proteins): gene, target, $\hat{\beta}_{\text{OLS}}$, 95% CI, p.
  File: `MAL_omics/mal_risk_score_weights.csv`
- **S3 — Per-protein mediation** (378 proteins in the PRS, individual mediation for each): NDE, NIE, total, proportion mediated, NIE p.
  File: `MAL_omics/mal_mediation_supp.csv`

  Top 6 individual mediators (NIE $p < 0.05$): TREM2 (13%, $p = 0.017$), ADAMTS5 (5.7%, $p = 0.021$), SEZ6L (17%, $p = 0.025$), RSPO1 (14%, $p = 0.038$), GREM2 (12%, $p = 0.039$), ADH7 (7%, $p = 0.044$). All small in absolute magnitude.

Supplementary Figures

- **Supp Figure A — Forest plot: proteins $\sim$ MAL.** Top 30 FDR-significant proteins, $\hat{\beta}$ for MAL association with 95% CI. File: `MAL_omics/SuppFig_forest_MAL.png`
- **Supp Figure B — Forest plot: proteins $\sim$ emphysema progression.** Same 30 proteins, $\hat{\beta}$ for progression association with 95% CI (adjusted for baseline lung density and covariates). File: `MAL_omics/SuppFig_forest_progression.png`

What Was Not Found / Assumptions Made

1. **$\mathbf{FEV_1}$% predicted**: column `FEV1pp_P2` does not exist in the phenotype file. Not included in Table 1 or any model. Check the data dictionary for the correct column name.
2. **BMI**: no BMI column found. Not included in Table 1 or the clinical regression model. The clinical model therefore adjusts for age, sex, race, smoking, and pack-years only. AIC comparisons may change once BMI is added.
3. **Figure 1 (study schematic)**: not automatically generated. Needs to be drawn manually.
4. **STRING network PDF**: only PNG was saved. The `STRINGdb` R package does not support direct PDF output via `plot_network()`; a PNG at 150 dpi was used instead.
5. **Table 1 values**: the exact means and percentages are in the notebook HTML output (cell 9) but could not be embedded here because the data are PHI and not committed. The LaTeX rows are placeholders.